IMPLEMENTASI & PENGUJIAN PERANGKAT LUNAK

TUGAS 2 – SEGITIGA & ANALISIS LAYANAN *858# dsb



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SEGITIGA

1. KODE PROGRAM SEGITIGA

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b, c;

    cout << "=== PROGRAM MENENTUKAN JENIS SEGITIGA ===" << endl;

    cout << "Masukkan sisi pertama  : ";
    cin >> a;

    cout << "Masukkan sisi kedua    : ";
    cin >> b;

    cout << "Masukkan sisi ketiga   : ";
    cin >> c;

    // Mengecek apakah dapat membentuk segitiga
    if (a <= 0 || b <= 0 || c <= 0) {
        cout << "Bukan segitiga" << endl;
    }
    else if (a + b <= c || a + c <= b || b + c <= a) {
        cout << "Bukan segitiga" << endl;
    }

    // Segitiga sama sisi
    else if (a == b && b == c) {
        cout << "Segitiga sama sisi" << endl;
    }

    // Segitiga siku-siku
    else if (a * a + b * b == c * c ||
             a * a + c * c == b * b ||
             b * b + c * c == a * a) {
        cout << "Segitiga siku-siku" << endl;
    }

    // Segitiga sama kaki
    else if (a == b || a == c || b == c) {
        cout << "Segitiga sama kaki" << endl;
    }

    // Segitiga sembarang
    else {
        cout << "Segitiga sembarang" << endl;
    }

    return 0;
}
```

2. OUTPUT

2.1 SEGITIGA SEMBARANG

```
=== PROGRAM MENENTUKAN JENIS SEGITIGA ===
Masukkan sisi pertama  : 4
Masukkan sisi kedua    : 5
Masukkan sisi ketiga   : 6
Segitiga sembarang

Process returned 0 (0x0)   execution time : 1
72.379 s
Press any key to continue.
```

2.2 SEGITIGA SAMA SISI

```
=== PROGRAM MENENTUKAN JENIS SEGITIGA ===
Masukkan sisi pertama  : 5
Masukkan sisi kedua    : 5
Masukkan sisi ketiga   : 5
Segitiga sama sisi

Process returned 0 (0x0)   execution time :
Press any key to continue.
```

2.3 SEGITIGA SIKU-SIKU

```
=== PROGRAM MENENTUKAN JENIS SEGITIGA ===
Masukkan sisi pertama  : 3
Masukkan sisi kedua    : 4
Masukkan sisi ketiga   : 5
Segitiga siku-siku

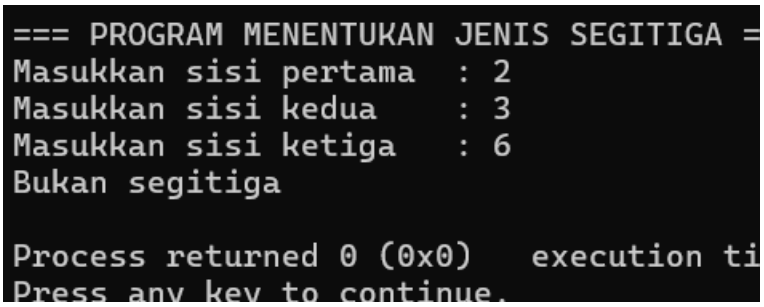
Process returned 0 (0x0)   execution time : 5
Press any key to continue.
```

2.4 SEGITIGA SAMA KAKI

```
=== PROGRAM MENENTUKAN JENIS SEGITIGA ===
Masukkan sisi pertama  : 5
Masukkan sisi kedua    : 5
Masukkan sisi ketiga   : 7
Segitiga sama kaki

Process returned 0 (0x0)   execution time
Press any key to continue.
```

2.5 BUKAN SEGITIGA



LAYANAN *858#

1. DFD LAYANAN *858#

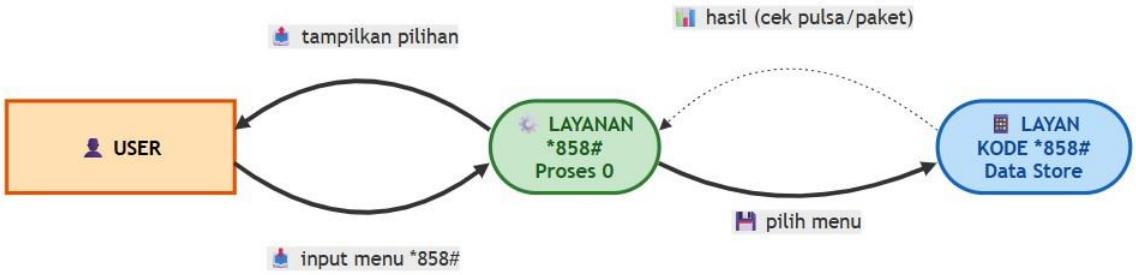
1.1 LEVEL 0 (CONTEXT DIAGRAM)

Entitas eksternal: User

Proses utama: **Sistem 858#*

Alur:

- User → input menu (*858#)
- Sistem → tampilkan pilihan
- User → pilih menu
- Sistem → hasil (cek pulsa / paket / dll)



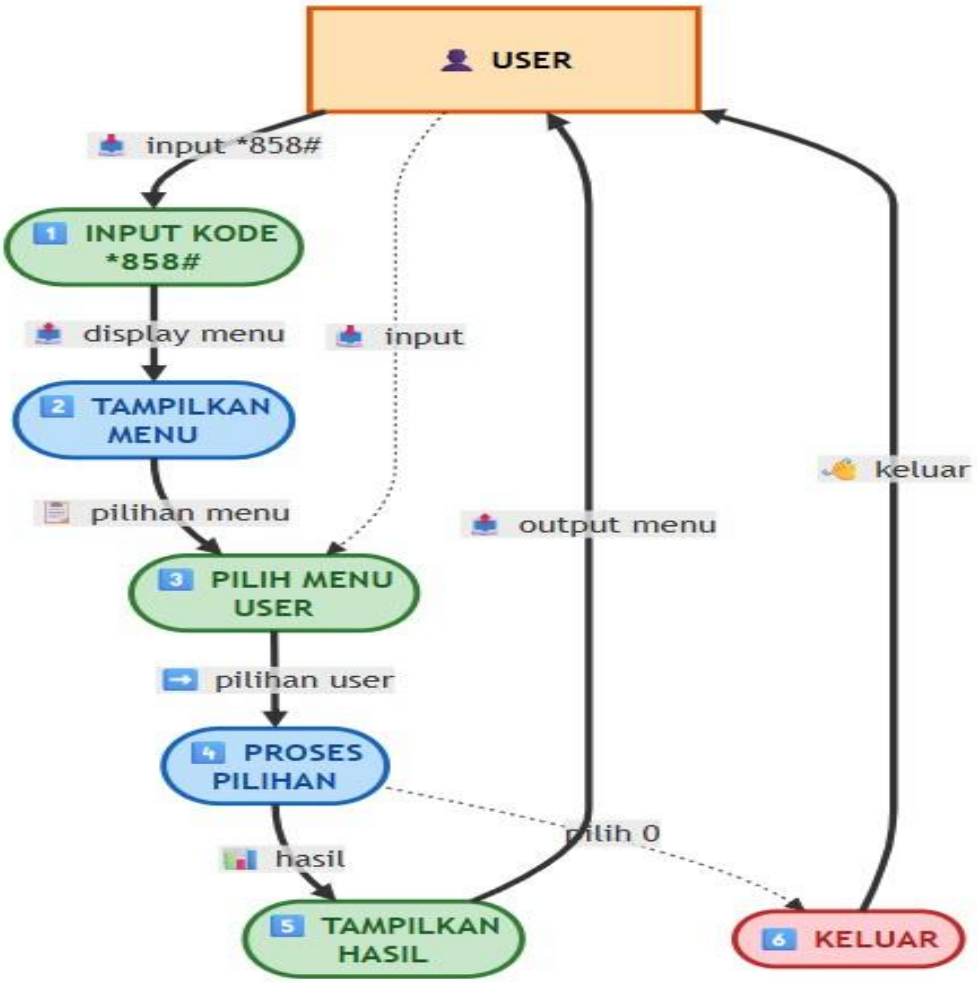
1.2 LEVEL 1 (DETAIL PROSES)

Proses utama dipecah jadi:

1. Input kode *858#
2. Tampilkan menu
3. Pilih menu user
4. Proses pilihan
5. Tampilkan hasil
6. Keluar

Contoh alur:

- User → *858#
- Sistem → Menu:
    1. Cek Pulsa
    2. Paket Internet
    3. Promo
    4. Info Bantuan
- User pilih → misal 1
- Sistem → tampilkan pulsa



2. PSPEC

2.1 CONTOH JALUR: TRANSFER PULSA

BEGIN

DISPLAY "Menu Transfer Pulsa"

INPUT nomor tujuan

DISPLAY "Masukkan nominal pulsa"

INPUT nominal pulsa

DISPLAY nomor tujuan dan nominal pulsa

DISPLAY "Apakah data sudah benar? (Y/N)"

INPUT konfirmasi

IF konfirmasi = "Y" OR konfirmasi = "y" THEN

PROCESS transfer pulsa

DISPLAY "Transfer pulsa berhasil"

ELSE

DISPLAY "Transfer pulsa dibatalkan"

ENDIF

END

2.2 CONTOH JALUR: CEK PULSA

BEGIN

DISPLAY "Menu Cek Pulsa"

PROCESS mengambil informasi pulsa pengguna

DISPLAY jumlah pulsa pengguna

DISPLAY "Cek pulsa selesai"

END

2.3 CONTOH JALUR: PAKET INTERNET

BEGIN

DISPLAY "Menu Paket Internet"

DISPLAY daftar paket internet

INPUT pilihan paket

IF pilihan paket tersedia THEN

DISPLAY harga paket

DISPLAY "Apakah ingin membeli paket? (Y/N)"

INPUT konfirmasi

IF konfirmasi = "Y" OR konfirmasi = "y" THEN

PROCESS pengecekan saldo

PROCESS pembelian paket

DISPLAY "Paket internet berhasil dibeli"

ELSE

```
        DISPLAY "Pembelian paket dibatalkan"
    ENDIF
ELSE
    DISPLAY "Paket tidak tersedia"
ENDIF
END
```

2.4 CONTOH JALUR: PROMO

```
BEGIN
DISPLAY "Menu Promo"
PROCESS mengambil informasi promo
IF promo tersedia THEN
    DISPLAY daftar promo
ELSE
    DISPLAY "Tidak ada promo yang tersedia"
ENDIF
END
```

2.5 CONTOH JALUR: INFO BANTUAN

```
BEGIN
DISPLAY "Menu Info Bantuan"
DISPLAY informasi layanan bantuan
DISPLAY nomor Customer Service
DISPLAY informasi bantuan lainnya
END
```

3. KODE PROGRAM LAYANAN *858#

```cpp
#include <iostream>
using namespace std;

int main() {
    int pilihan;
    int pilihanPaket;
    string nomorTujuan;
    int nominal;
    char konfirmasi;

    cout << "================================" << endl;
    cout << "          LAYANAN *858#" << endl;
    cout << "================================" << endl;
    cout << "1. Transfer Pulsa" << endl;
    cout << "2. Cek Pulsa" << endl;
    cout << "3. Paket Internet" << endl;
    cout << "4. Promo" << endl;
    cout << "5. Info Bantuan" << endl;
    cout << "================================" << endl;

    cout << "Pilih menu: ";
    cin >> pilihan;

    // MENU 1 - TRANSFER PULSA
    if (pilihan == 1) {

        cout << endl;
        cout << "================================" << endl;
        cout << "         TRANSFER PULSA" << endl;
        cout << "================================" << endl;

        cout << "Masukkan nomor tujuan : ";
        cin >> nomorTujuan;

        cout << "Masukkan nominal pulsa : Rp";
        cin >> nominal;

        cout << endl;
        cout << "Nomor tujuan : " << nomorTujuan << endl;
        cout << "Nominal      : Rp" << nominal << endl;

        cout << "Apakah data sudah benar? (Y/N): ";
        cin >> konfirmasi;

        if (konfirmasi == 'Y' || konfirmasi == 'y') {
            cout << endl;
            cout << "Memproses transfer pulsa..." << endl;
            cout << "Transfer pulsa berhasil!" << endl;
            cout << "Pulsa Rp" << nominal
                 << " berhasil dikirim ke "
                 << nomorTujuan << "." << endl;
        }
        else {
            cout << endl;
            cout << "Transfer pulsa dibatalkan." << endl;
```

```
            cout << "Transfer pulsa dibatalkan." << endl;
        }
    }

    // MENU 2 - CEK PULSA
    else if (pilihan == 2) {

        cout << endl;
        cout << "================================" << endl;
        cout << "             CEK PULSA" << endl;
        cout << "================================" << endl;

        cout << "Pulsa Anda saat ini : Rp50.000" << endl;
        cout << "Cek pulsa berhasil." << endl;
    }

    // MENU 3 - PAKET INTERNET
    else if (pilihan == 3) {

        cout << endl;
        cout << "================================" << endl;
        cout << "          PAKET INTERNET" << endl;
        cout << "================================" << endl;

        cout << "1. Paket 5 GB  - Rp25.000" << endl;
        cout << "2. Paket 10 GB - Rp45.000" << endl;
        cout << "3. Paket 15 GB - Rp60.000" << endl;

        cout << endl;
        cout << "Pilih paket: ";
        cin >> pilihanPaket;

        if (pilihanPaket == 1) {

            cout << endl;
            cout << "Anda memilih Paket 5 GB." << endl;
            cout << "Harga paket: Rp25.000" << endl;

            cout << "Apakah ingin membeli paket? (Y/N): ";
            cin >> konfirmasi;

            if (konfirmasi == 'Y' || konfirmasi == 'y') {
                cout << "Pembelian berhasil!" << endl;
                cout << "Paket 5 GB telah aktif." << endl;
            }
            else {
                cout << "Pembelian paket dibatalkan." << endl;
            }
        }

        else if (pilihanPaket == 2) {

            cout << endl;
            cout << "Anda memilih Paket 10 GB." << endl;
            cout << "Harga paket: Rp45.000" << endl;
```

```
            cout << "Apakah ingin membeli paket? (Y/N): ";
            cin >> konfirmasi;

            if (konfirmasi == 'Y' || konfirmasi == 'y') {
                cout << "Pembelian berhasil!" << endl;
                cout << "Paket 10 GB telah aktif." << endl;
            }
            else {
                cout << "Pembelian paket dibatalkan." << endl;
            }
        }

        else if (pilihanPaket == 3) {

            cout << endl;
            cout << "Anda memilih Paket 15 GB." << endl;
            cout << "Harga paket: Rp60.000" << endl;

            cout << "Apakah ingin membeli paket? (Y/N): ";
            cin >> konfirmasi;

            if (konfirmasi == 'Y' || konfirmasi == 'y') {
                cout << "Pembelian berhasil!" << endl;
                cout << "Paket 15 GB telah aktif." << endl;
            }
            else {
                cout << "Pembelian paket dibatalkan." << endl;
            }
        }

        else {
            cout << "Pilihan paket tidak tersedia." << endl;
        }
    }

    // MENU 4 - PROMO
    else if (pilihan == 4) {

        cout << endl;
        cout << "================================" << endl;
        cout << "               PROMO" << endl;
        cout << "================================" << endl;

        cout << "1. Promo Kuota Hemat" << endl;
        cout << "2. Promo Nelpon" << endl;
        cout << "3. Promo Internet" << endl;

        cout << endl;
        cout << "Promo tersedia untuk pelanggan." << endl;
    }

    // MENU 5 - INFO BANTUAN
    else if (pilihan == 5) {

        cout << endl;
        cout << "================================" << endl;
        cout << "          INFO BANTUAN" << endl;
        cout << "================================" << endl;

        cout << "Customer Service : 188" << endl;
```

```
173         }
174
175         // PILIHAN TIDAK TERSEDIA
176         else {
177
178             cout << endl;
179             cout << "Pilihan menu tidak tersedia." << endl;
180         }
181
182         cout << endl;
183         cout << "==================================" << endl
184         cout << "           PROGRAM SELESAI" << endl;
185         cout << "==================================" << endl
186
187         return 0;
188     }
```

4. OUTPUT KODE LAYANAN *858#

4.1 TRANSFER PULSA

```
================================
1. Transfer Pulsa
2. Cek Pulsa
3. Paket Internet
4. Promo
5. Info Bantuan
================================
Pilih menu: 1

================================
         TRANSFER PULSA
================================
Masukkan nomor tujuan : 0895602509876
Masukkan nominal pulsa : Rp100000

Nomor tujuan : 0895602509876
Nominal      : Rp100000
Apakah data sudah benar? (Y/N): y

Memproses transfer pulsa...
Transfer pulsa berhasil!
Pulsa Rp100000 berhasil dikirim ke 0895602509876.
```

4.2 CEK PULSA

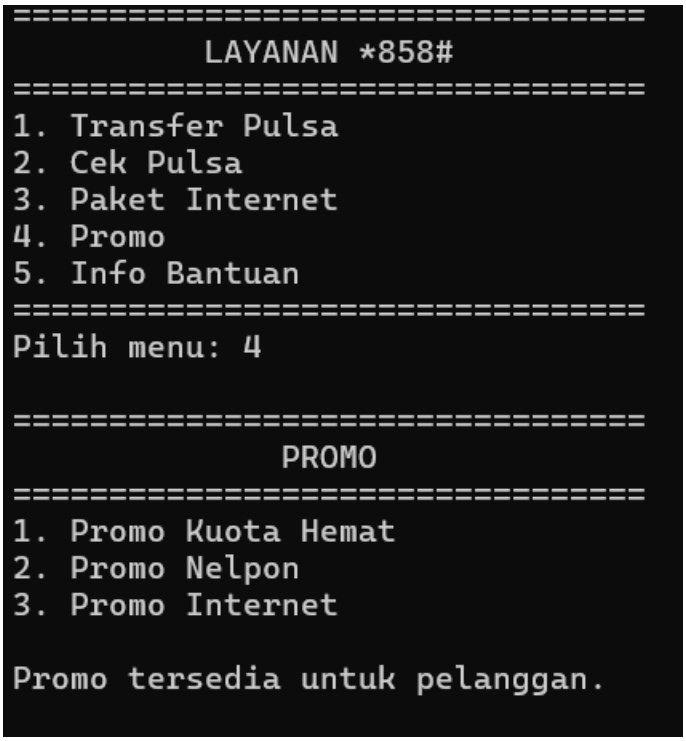
```
================================
          LAYANAN *858#
================================
1. Transfer Pulsa
2. Cek Pulsa
3. Paket Internet
4. Promo
5. Info Bantuan
================================
Pilih menu: 2

================================
             CEK PULSA
================================
Pulsa Anda saat ini : Rp50.000
Cek pulsa berhasil.
```

4.3 PAKET INTERNET



4.4 PROMO



4.5 INFO BANTUAN

